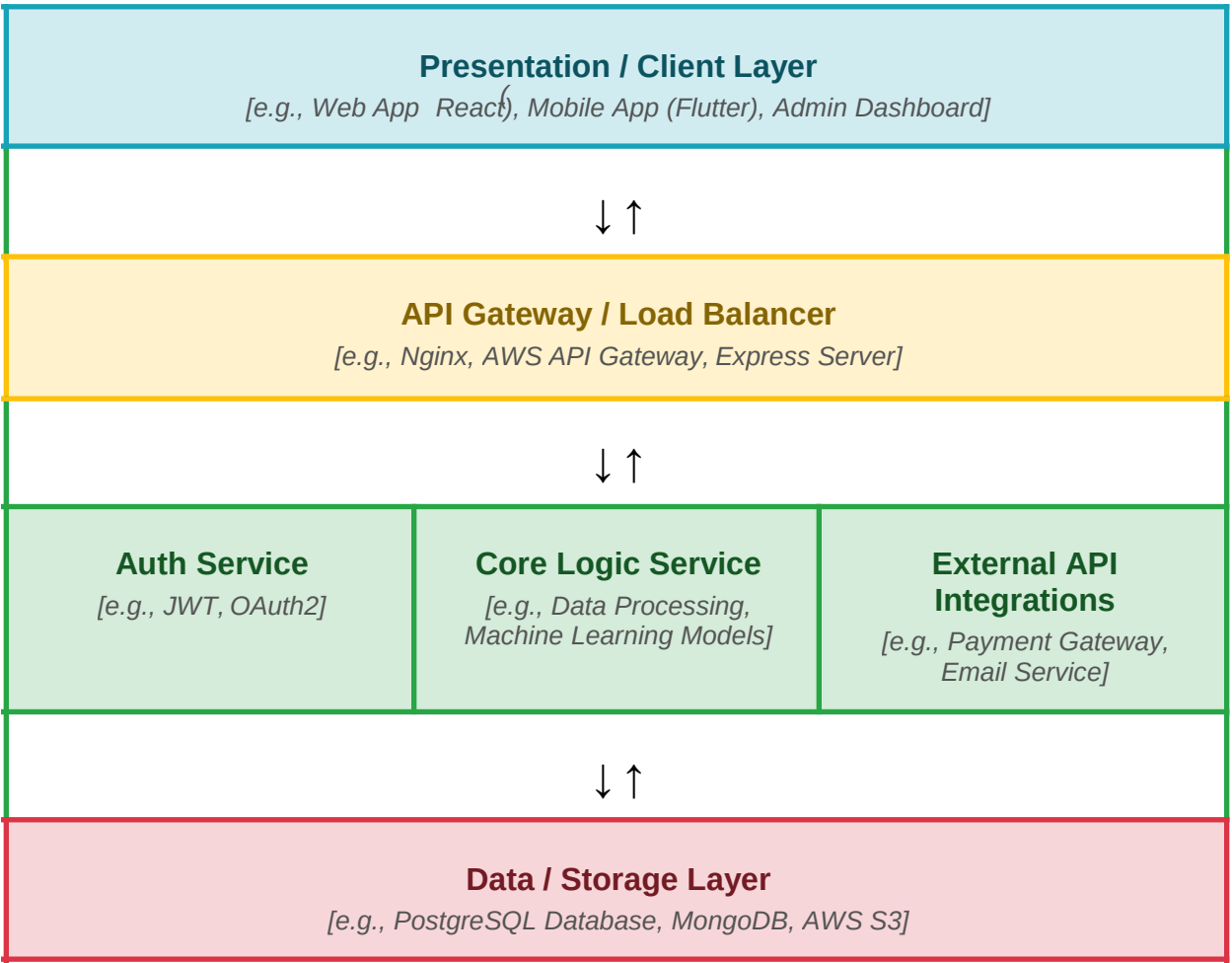


Solution Architecture

Date	12 Jun 2026
Team ID	xxxxxxx
Project Name	Smart Crop
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides interactive, responsive browser dashboards tailored to end-users. Enables farmers to submit 7 unique indicators (N, P, K, pH, temperature, humidity, rainfall) and visualizes regional agronomical data graphs for policymakers.	React.js, Tailwind CSS
API Gateway	Acts as a central entry system to manage, secure, forward, and balance incoming application traffic routes seamlessly down to specific computational microservices.	AWS API Gateway / Nginx
Auth Service	Manages secure login flows and handles token-based role definitions (RBAC). Restricts editing capabilities to specific datasets depending on user profiles.	OAuth2 / JWT Tokens
Core Logic Service	Hosts the target machine learning recommendation algorithms.	Python, FastAPI, scikit-learn

	<i>Evaluates environmental factors, computes crop suitability percentages, and generates crop ranking scores across all three defined user scenarios.</i>	
<i>External API Integrations</i>	<i>Pulls live atmospheric telemetry matrices and regional weather history parameters to provide instant situational evaluation baselines.</i>	<i>OpenWeatherMap API</i>
<i>Database Layer</i>	<i>Stores crop reference thresholds, structural compatibility tables, user profile settings, and persistent regional logging metrics.</i>	<i>PostgreSQL Relational DB</i>